

Packets and Frames :-

What are packets and Frames?

* Packets and Frames are small pieces of data, when forming together makes a larger piece of information or message. They are different things in OSI model.

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* A Packet is piece of data from layer 3 (Network layer) of OSI model. Containing information such as an IP header and payload. • A Frame however is used at layer 2 (Data link) of the OSI model, which encapsulates packets and adds additional information such as MAC addresses.

Note:- Its safe to assume that when we are talking about IP address we are talking about packets. when we talk about encapsulation we talk about Frames.

* Packets are efficient way of communicating data across networked devices such as those explained in Task 1. Because data is exchange in small pieces, there is less chance of bottlenecking, then large messages being sent once.

* Network is full of Standards and protocols that acts as set of rules and how packet is handled on a device. With billions of devices connected on Internet, things can quickly break down if there is no standardisation.

Ex:- Internet protocol. A packet using this protocol will have lot of headers that contains additional piece of information to data.

that is being sent across network. ✓ Internet Protocol header this all

Header	Description
Time to live	This field sets an expiry timer for packet to not clog up your network if it never manages to reach a host or escape!
Checksum	This field provides integrity checking for protocols such as TCP/IP. If any data is changed, this value will be different from what was expected and therefore corrupt.
Source Address	The IP Address of device that packet is being sent from so that data knows where to return to.
Destination Address	The device's IP address the packet is being sent to so that data knows where to travel next.

Quiz: -

1) What is the name of piece of data when it does have IP addressing information?

A) Packet

2) What is name of piece of data when it does not have IP addressing information?

A) Frame

Task 2: - TCP/IP (three way Handshake): -

* TCP is another one of these rules used in networking.

* This protocol is similar to OSI model that we previously discussed from 3 of this model.

* TCP/IP consists of 4 layers and is arguably just a summarised version of OSI model. These layers are

* Application * Transport * Internet * Network Interface

* Same as OSI model works, information is added to each layer of TCP model as piece of data (or packets) traverse it. As you

may recall this process is known as encapsulation. where reverse of this process is known as decapsulation.

* One defining feature of TCP is that connection based, which means TCP must establish connection b/w both client and device acting as server before data is sent.

* Because of this TCP guarantees that any data sent will be received on other end. This process is named as **Three Way handshake**.

* We discussed about advantages and disadvantages of TCP before.

* TCP packets contains various section of information known as headers that are added from encapsulation. let's explain some crucial headers in table below.

Header	Description
Source port	This value is port opened by sender to send the TCP packet from. This value is choose randomly (0-65535 that aren't already in use at the time)
Destination port	This value is port number than an application or service is running on remote host, Ex:- a web browser running on port 80. unlike source port, this value is not chosen at random
Source IP	This is IP address of device that is sending the packet
Destination IP	This is IP address of packet that is destined for
Sequence Number	When a connection occurs, the first piece of data transmitted is given a random number. we'll explain this more depth in further.
Acknowledgment Number	After piece of data has been given a sequence number, the number of next piece of data will have Sequence number + 1
Checksum	This value is what gives TCP integrity. A mathematical calculation is made where output is remembered. when receiving device performs mathematical calculations, the data must align if output is different from what was sent
Data	This header is where the data, i.e bytes of file that is being transmitted. is stored

Flag

This header determines how packet should be handled by either device during the handshake process. Specific flag will determine specific behaviours, which is what we'll discuss explain below.